CUSTOMER ANALYTICS / PYTHON PROJECT

From Transactions to Customer Strategy

Who drives value, who needs attention—and what should we do next?

1,000 transactions • 946 customers • 11 Apr–10 Jun 2023

Python • Pandas • RFM scoring • Customer insight

The business question is bigger than a segment label

RFM is useful only when it changes where we invest attention, offers and retention effort.

PROTECT

Which customers create disproportionate value—and how do we keep them?

RECOVER

Which customers may be drifting away, and what is a sensible win-back test?

GROW

Where can relevant product recommendations or better timing increase engagement?

Our approach Use transaction history to score Recency, Frequency and Monetary value; group customers; compare segment value and behaviour; then turn the strongest patterns into testable actions.

Important distinction: these are descriptive signals—not proof that a campaign will cause a lift.

What the data can—and cannot—tell us

TRANSACTIONS

1,000

Unique orders

CUSTOMERS

946

Unique customer IDs

OBSERVED REVENUE

513.7k

Currency not specified in source data

COVERAGE

61 days

11 Apr–10 Jun 2023

Available

- Customer, order, date and transaction amount
- Product label and location
- Observed repeat purchase and spend

Not available

- Returns, reviews or satisfaction scores
- Detailed product attributes or basket combinations
- A full year of data for seasonal comparisons

Data checks: 0 missing cells • 0 exact duplicate rows

RFM gives each customer three complementary lenses

R

RECENCY

How recently did they buy?

Lower days = stronger recent activity

F

FREQUENCY

How often did they buy?

More distinct orders = stronger repeat behaviour

M

MONETARY

How much did they spend?

Higher observed spend = greater revenue value

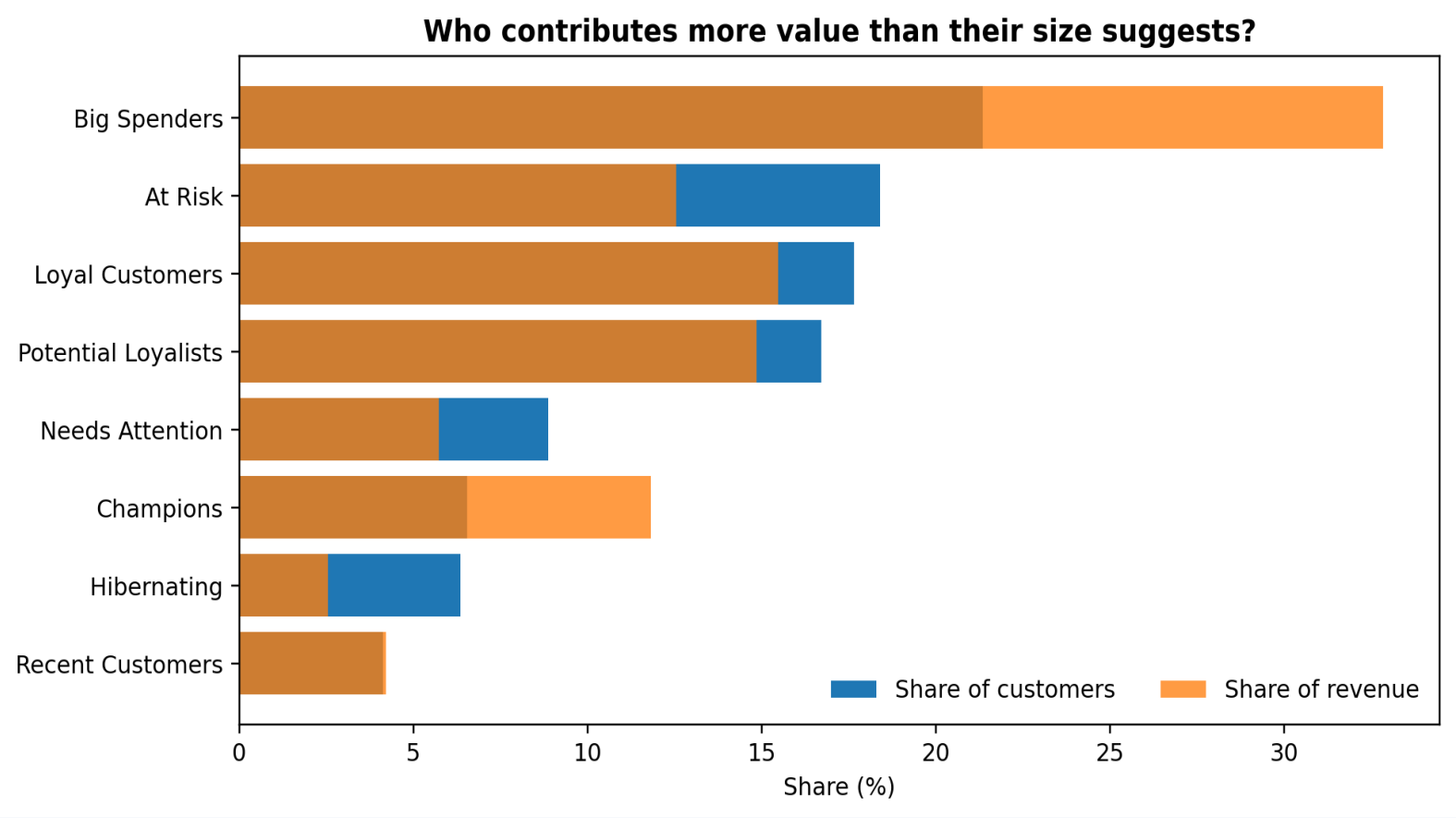
Scoring logic

Each metric is ranked into five quantile-based scores; transparent rules then assign one of eight customer segments.

Reference date: 11 Jun 2023 (the day after the final observed transaction). Segment labels depend on the chosen rules and period.

The first insight: customer share and revenue share are not the same

Comparing both shares helps identify segments worth protecting—not just the biggest groups.



BIG SPENDERS

32.8% of revenue from 21.4% of customers.

CHAMPIONS

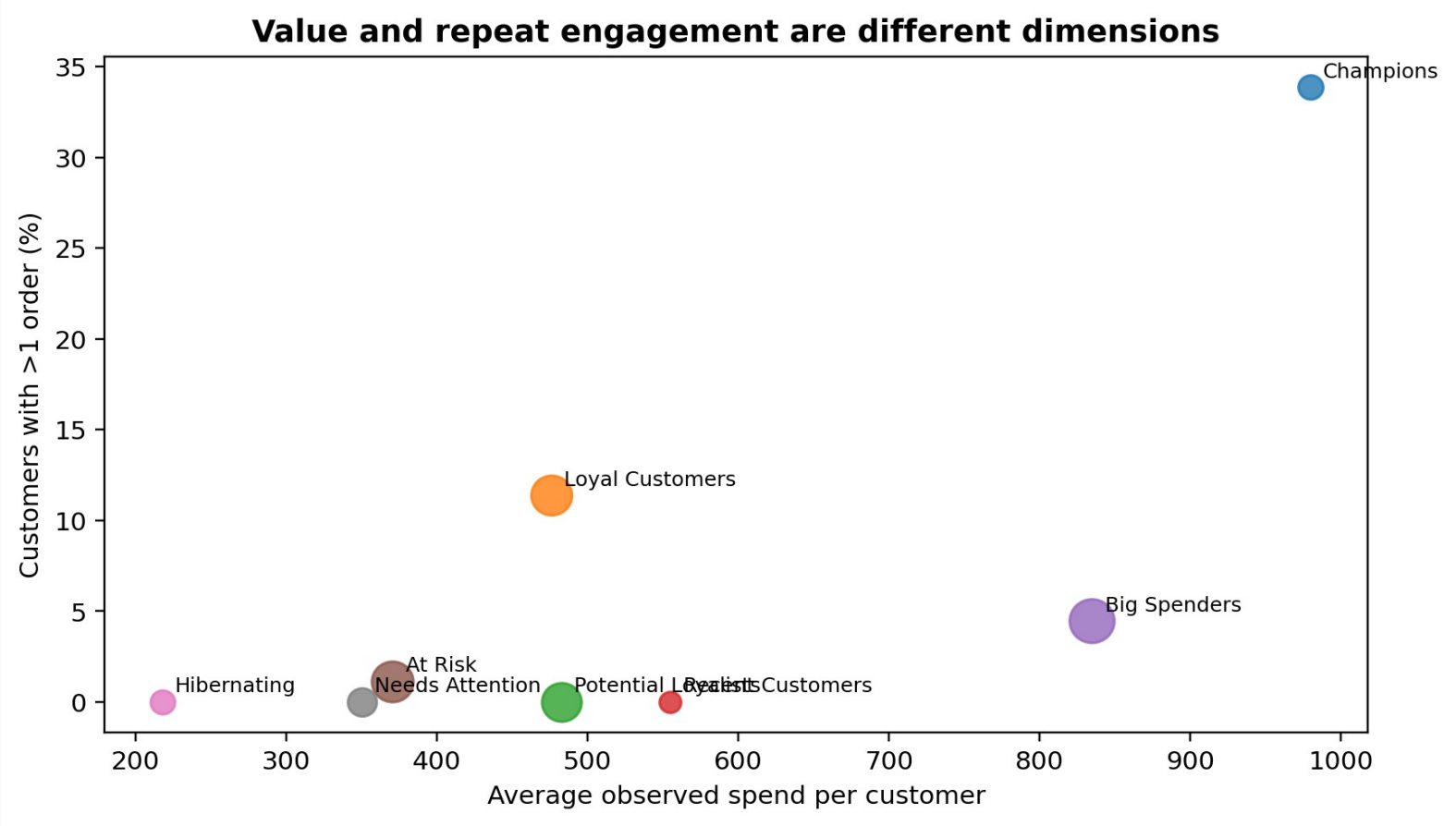
11.8% of revenue; 979 average observed spend per customer.

BUSINESS IMPLICATION

Protect high-value experiences while testing how to grow repeat behaviour.

The second insight: high spend does not automatically mean loyalty

Repeat purchasing is a useful behavioural proxy, but not a direct measure of satisfaction.



CHAMPIONS

33.9% have placed more than one observed order.

BIG SPENDERS

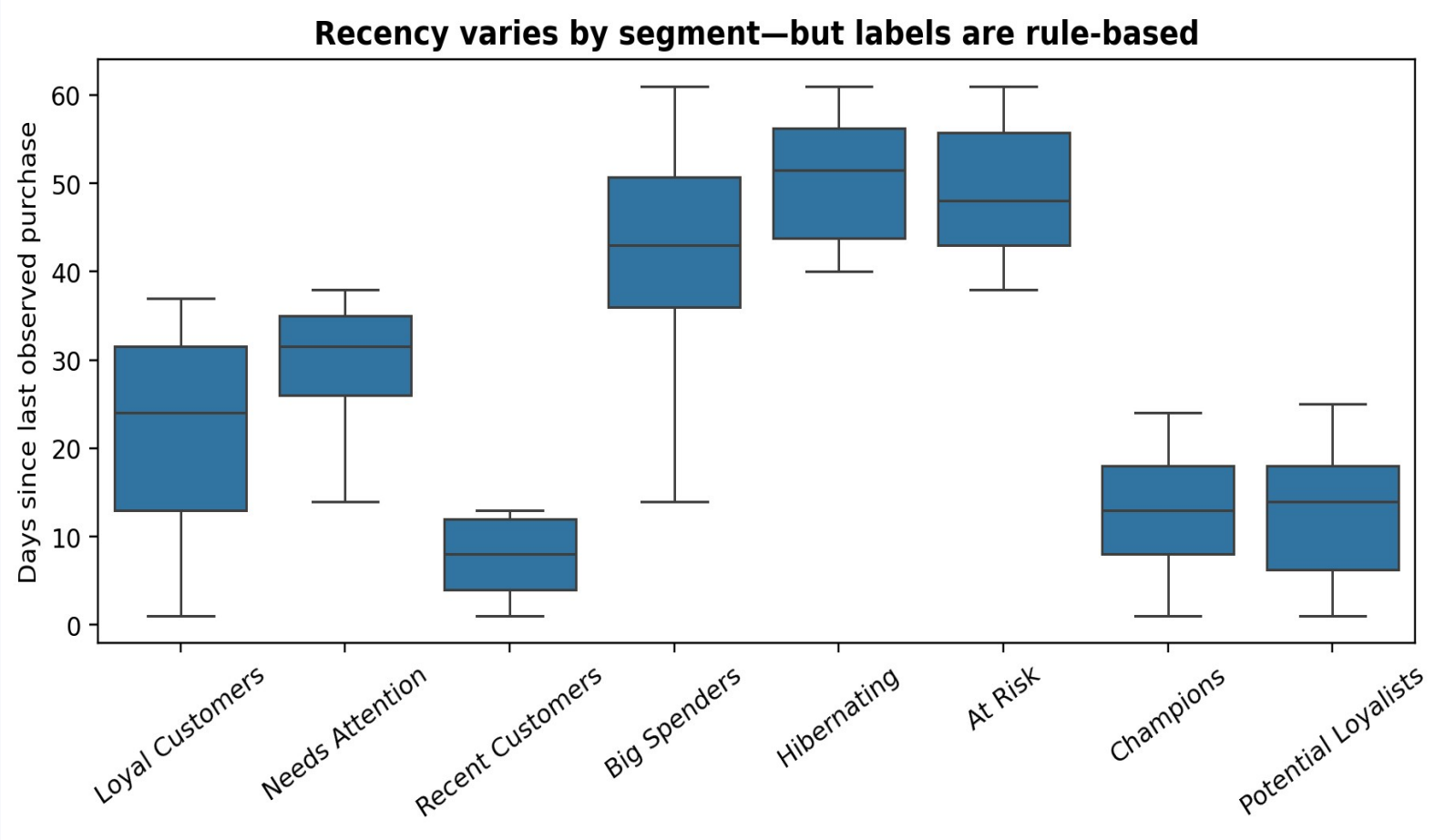
4.5% repeat-purchase rate in this observed period.

WHAT TO DO

Reward repeat behaviour; separately monitor high-spend customers with low repeat activity.

Where should retention effort go first?

Use recency together with frequency and spend; do not treat a segment label as a guaranteed churn prediction.



AT RISK

174 customers; average observed spend 371.

HIBERNATING

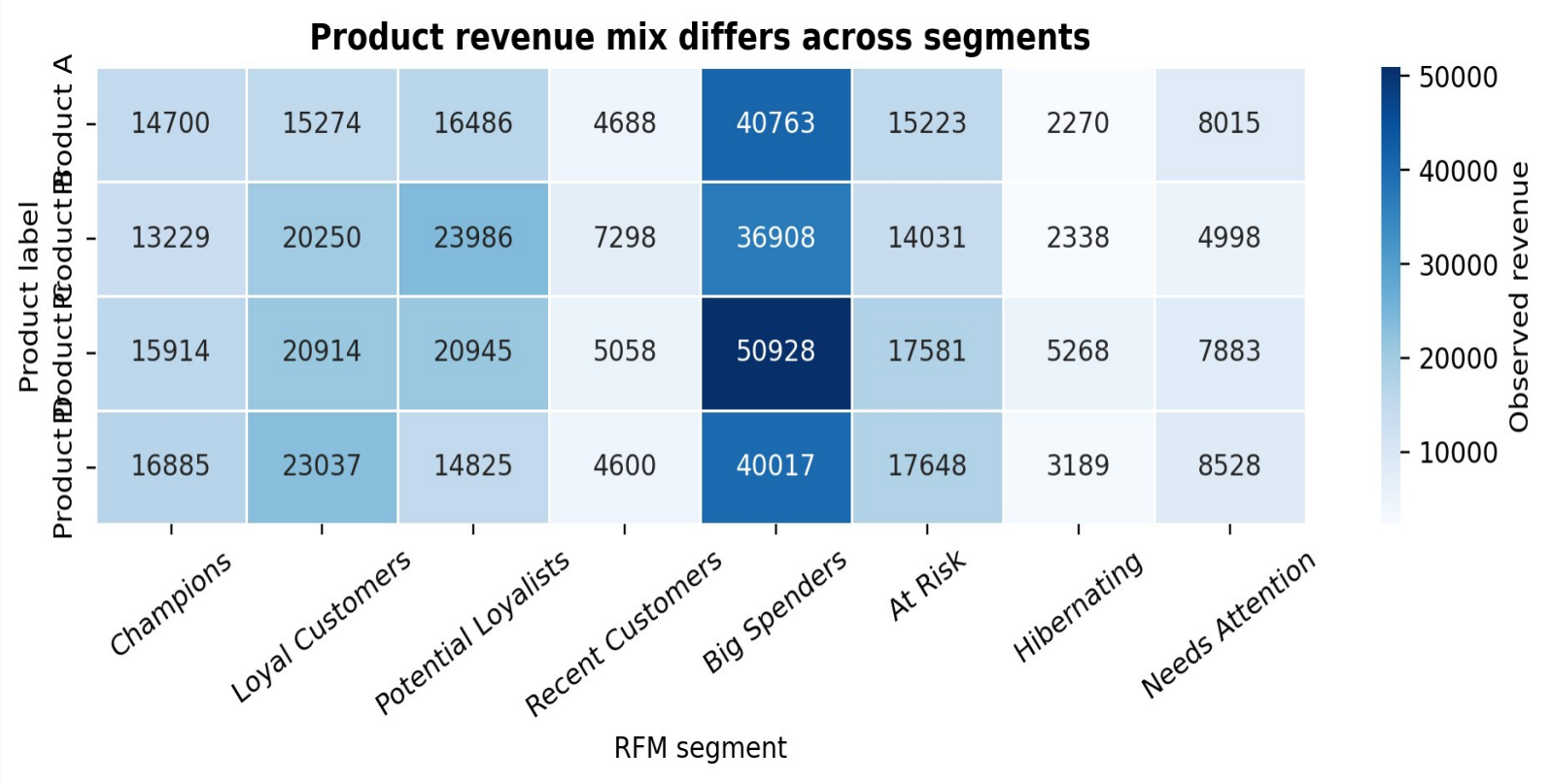
60 customers; average observed spend 218.

RECOMMENDED TEST

Prioritise high-value, less-recent customers; test personalised reminders and controlled incentives.

Product patterns can shape recommendations—but cannot prove bundles

The product labels are generic, so findings indicate where to investigate rather than why customers buy.



OBSERVED SIGNAL

Product C leads observed revenue among Big Spenders (50,928).

WHAT WE CAN'T CLAIM

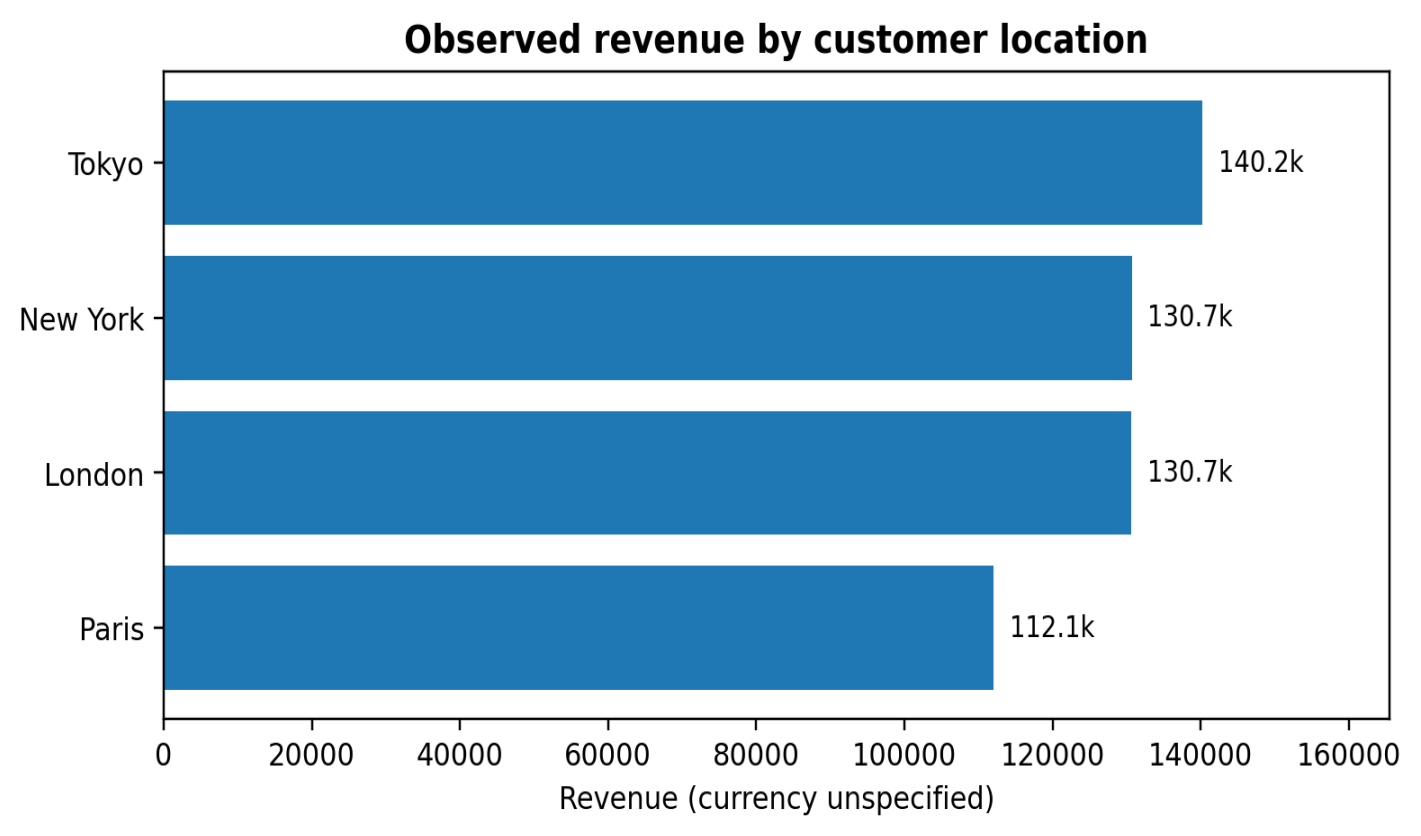
No orders contain multiple distinct product labels; basket association rules cannot be calculated from this file.

NEXT STEP

Test recommendations as hypotheses and measure conversion / incremental revenue.

Geography helps prioritise tests—not explain customer behaviour

Compare total revenue with customer count and average revenue; avoid assuming location causes the difference.



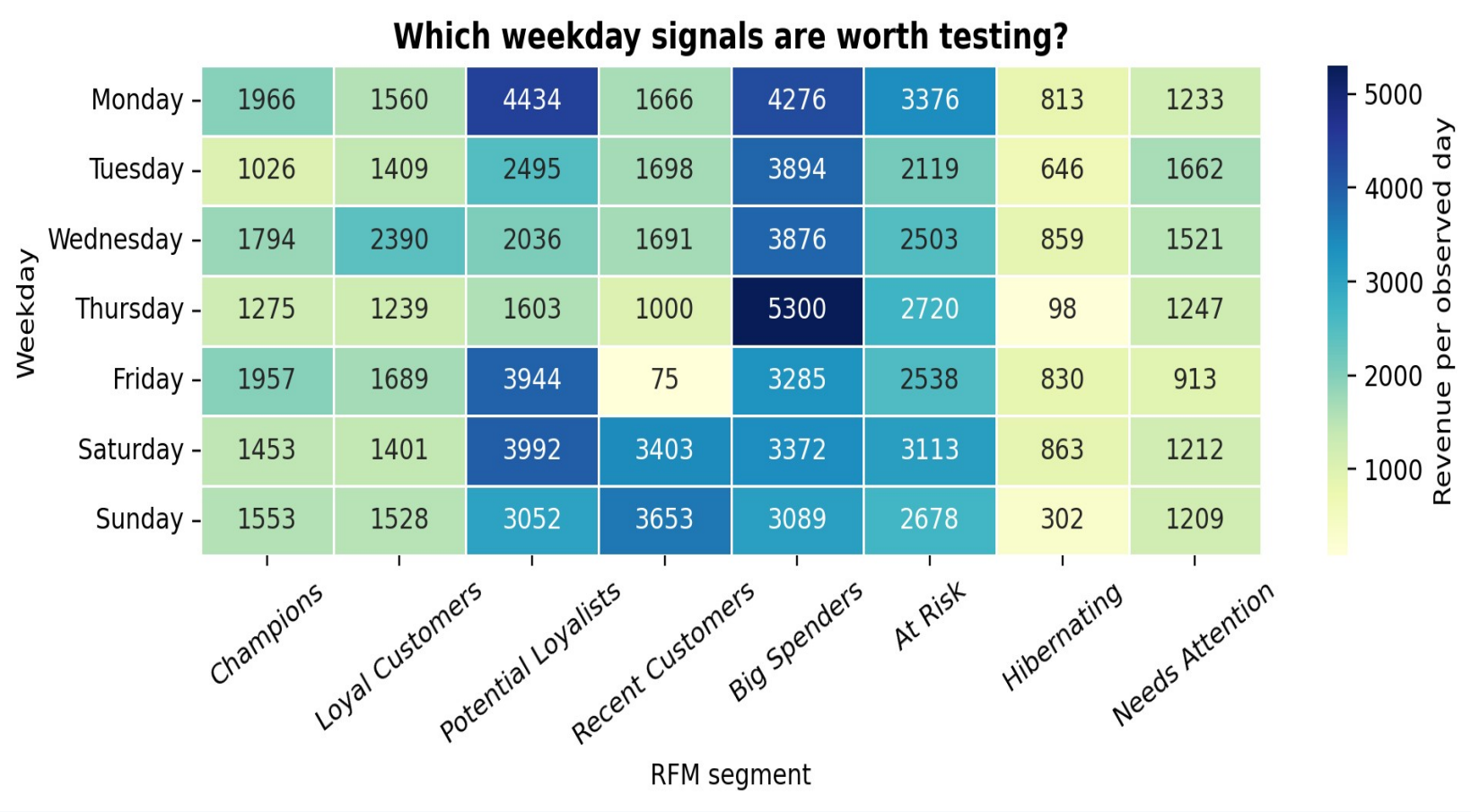
How to act on this

- Localise messaging only where the segment/product mix supports it.
- Run a small campaign or holdout test before scaling.
- Compare conversion and revenue per customer—not just total sales.

Customer location is assigned from the most frequently observed transaction location.

Timing differences suggest experiments, not a calendar rule

Daily revenue is normalised by observed active dates to reduce distortion from unequal coverage.



Use peaks as test ideas

- Champions: Monday
- Loyal Customers: Wednesday
- Potential Loyalists: Monday
- Big Spenders: Thursday

Short observation window; compare results over a longer period before changing send times or staffing.

A practical 30-day action plan

Move from descriptive insight to controlled tests with clear success measures.

1	WEEK 1	**Prioritise**	Rank At Risk / Hibernating customers by historical spend and recency; define eligible cohorts.
2	WEEK 2	**Launch tests**	Test personalised win-back vs. standard messaging; keep a holdout group.
3	WEEK 3	**Grow value**	Test segment-specific product recommendations and campaign timing.
4	WEEK 4	**Evaluate**	Compare repeat purchase, conversion, incremental revenue and cost per reactivated customer.

Success means incremental lift versus a control group—not simply clicks or revenue after a campaign.

The next dataset should close the biggest decision gaps

No returns / feedback

Satisfaction and dissatisfaction are not directly measurable.

COLLECT

Returns, reason codes, ratings, complaints

One product label per order

Cannot infer products bought together.

COLLECT

Order-line / basket-level product IDs

~2 months of history

Annual seasonality and long-term churn are not established.

COLLECT

12+ months of dated transactions

Better data would let us move from descriptive segmentation toward retention measurement, product affinity and customer-experience diagnosis.

THE TAKEAWAY

Segment to prioritise. Experiment to learn. Measure to prove.

RFM provides a useful starting map of customer value and engagement. The next step is to test targeted actions and measure incremental impact.